📚 Respiratory System: Comprehensive Nursing Study Guide

🟪 Summary

📝 **Overview**: This guide synthesizes the mechanical and chemical failures of the respiratory system, integrating critical care concepts, post-operative management, and emergency protocols[1]. The core problem addressed is **Ineffective Gas Exchange** leading to systemic cell death[2]. The respiratory system functions to supply the body with oxygen (O2) and remove carbon dioxide (CO2), essential for homeostasis[3].

- 🩺 **Top 3 Priority Nursing Interventions**:
  - **Airway Patency**: Immediate action for obstruction (e.g., Head-Tilt/Chin-Lift, Suctioning) to ensure ventilation[4].
  - **Oxygenation Strategy**: Titrate O2 to maintain saturation >92% (or 88-92% for COPD) using the least invasive device first[5].
  - **Hemodynamic Support**: Manage cardiac consequences (e.g., Cor Pulmonale, Acidosis) and monitor MAP > 65 mmHg for organ perfusion[6].
- ⚡ **Key Warning Signs**:
  - **Agitation/Restlessness**: The #1 early sign of Hypoxia and Airway Obstruction[7].
  - **Stridor**: High-pitched sound indicating upper airway obstruction (Medical Emergency)[8].
  - **Silent Chest**: Absence of wheezing in an asthmatic, signaling impending arrest[9].
  - **Tracheal Deviation**: Late sign of Tension Pneumothorax[10].
- 🩺 **Overall Goal of Care**: To match Ventilation (V) with Perfusion (Q), prevent irreversible organ damage, and restore baseline function[11].

🟪 I. Anatomy & Physiology

🟦 A. Structural Mechanics

- **Epiglottis**: A flap preventing aspiration during swallowing[12].
- **Larynx**: Vital airway containing vocal folds; a primary landmark for intubation[13].
- **Trachea**: Narrowing here causes **Stridor**[14].
- **Carina**: The point where the trachea divides[15].
  - 🎯 **Right Bronchus**: Wider, shorter, and more vertical than the left[16].
  - 🟨 **Caution**: This makes the right bronchus the most common site for aspiration or Endotracheal (ET) tube misplacement[17].

- **Alveoli**: The primary site of gas exchange, wrapped in a capillary mesh[18].

🟦 B. Physiology of Gas Exchange

- **Ventilation**: Mechanical flow of air in/out of alveoli, driven by the diaphragm[19].
- **Diffusion**: Exchange of O2 and CO2 across the alveolar-capillary membrane[20].
- **Perfusion**: Transport of oxygenated blood to tissues[21].
- **V/Q Mismatch**: The primary cause of hypoxemia in Pulmonary Embolism (blockage of flow) or Pneumonia (blockage of air)[22].

🟪 II. Patient Assessment (The "Look, Listen, Feel")

🟦 Assessment Findings Table

Technique	Findings	Clinical Significance
Inspection	**Tripod Position**	Leaning forward to expand chest (COPD/Asthma)[23].
	Accessory Muscle Use	Sternocleidomastoid/Scalenes active indicates distress[24].
	Barrel Chest	Indicates air trapping/hyperinflation seen in Emphysema[25].
	Paradoxical Movement	🎯 Chest moves *in* on inspiration; hallmark of **Flail Chest**[26].
Auscultation	**Crackles (Rales)**	Fluid in alveoli (HF, Pneumonia, Pulmonary Edema)[27].

	Wheezing	Bronchoconstriction (Asthma/COPD)[28].
	Stridor	🚨 High-pitched upper airway obstruction (Medical Emergency)[29].
	Silent Chest	🚨 No air movement (Status Asthmaticus)[30].
Palpation	**Tracheal Deviation**	Shifts away from affected side in Tension Pneumothorax[31].
	Tactile Fremitus	Decreased in Pneumothorax/Effusion; Increased in Pneumonia[32].
	Crepitus	Subcutaneous emphysema (air trapped under skin)[33].

🟪 III. Arterial Blood Gas (ABG) Mastery

🟦 Normal Ranges 🧪

- **pH**: 7.35 – 7.45 [34]
- **PaCO2** (Respiratory Driver/Acid): 35 – 45 mm Hg [35]
- **HCO3** (Metabolic/Kidney/Base): 22 – 26 mEq/L [36]
- **PaO2**: 80 – 100 mm Hg [37]

🟦 Tic-Tac-Toe Interpretation Method

1. **Acidosis vs. Alkalosis**: pH < 7.35 is Acidosis; pH > 7.45 is Alkalosis[38].
2. **Respiratory Cause**: PaCO2 moves in the **Opposite** direction of pH (ROME)[39].
    - **Respiratory Acidosis**: pH Low, PaCO2 High (e.g., COPD, Opioid Overdose)[40].

- **Respiratory Alkalosis**: pH High, PaCO2 Low (e.g., Hyperventilation/Anxiety)[41].

3. **Metabolic Cause**: HCO3 moves in the **Same** direction as pH[42].
   - **Metabolic Acidosis**: pH Low, HCO3 Low (e.g., DKA, Renal Failure)[43].
   - **Metabolic Alkalosis**: pH High, HCO3 High (e.g., Vomiting, NG Suction)[44].

🟪 IV. Nursing Interventions & Procedures

🟦 Airway Clearance & Breathing Techniques

- 🩺 **Incentive Spirometer**: 10x/hour while awake to prevent atelectasis[45].
- 🩺 **Huff Coughing**: Clears secretions with less force than traditional coughing[46].
- 🩺 **Splinting**: Using a pillow to support abdominal incisions during coughing to prevent wound dehiscence[47].
- 🩺 **Pursed-Lip Breathing**: Prolongs exhalation to prevent bronchiolar collapse; gives patient control[48484848].

🟦 Oxygen Delivery Systems

System	Flow Rate	Notes
Nasal Cannula	1-6 L/min	Watch for skin breakdown[49].
Venturi Mask	Variable	🎯 Delivers precise FiO2. **Best for COPD patients**[50].
Non-Rebreather	10-15 L/min	Reservoir bag must stay inflated. For emergencies[51].

🟦 Chest Tubes

- **Purpose**: Remove air/fluid from pleural space[52].
- **Water Seal Chamber**:

- **Tidaling**: Water rises/falls with breath = **Normal**[53].
- **Continuous Bubbling**: Indicates an **Air Leak** (Check connections)[54].

- **Safety**:
  - 🚨 **Report**: Drainage > 100 mL/hr (Suggests Hemorrhage)[55].
  - **Dislodgement**: Cover site with sterile dressing taped on **3 sides**[56].

🟪 V. Comparative Pathology

🟦 Respiratory Failure: Type 1 vs. Type 2

Feature	Hypoxemic Failure (Type 1)	Hypercapnic Failure (Type 2)
Primary Defect	Oxygenation Failure [57]	Ventilatory Failure [58]
Criteria	🎯 PaO2 < 60 mm Hg [59]	🎯 PaCO2 > 50 mm Hg (with Acidosis) [60]
Mechanism	V/Q Mismatch or Shunt [61]	Reduced respiratory drive or chest wall failure [62]
S/S	Agitation, Cyanosis [63]	Somnolence, Morning Headache [64]

🟦 Post-Operative Complications Timeline

- 🟩 **Atelectasis (Collapse)**:
  - **Timing**: Early (Post-op Day 1-2)[65].
  - **Mechanism**: Alveolar collapse due to hypoventilation/immobility[66].
  - **Intervention**: TCDB, Early Ambulation[67].
- 🟩 **Pneumonia (Infection)**:
  - **Timing**: Later (Day 3+)[68].

- **Signs**: Productive cough, higher fever, consolidations[69].

🟪 VI. Critical Conditions

🟦 COPD (Chronic Obstructive Pulmonary Disease)

- **Pathophysiology**: Chronic inflammation, Emphysema (alveolar destruction), and Chronic Bronchitis[70].
- **Diet**: High calorie, high protein, small frequent meals; avoid high carbs (produces CO2)[71].
- 🟨 **Oxygen Caution**: In CO2 retainers, high O2 may suppress hypoxic drive. Target SpO2 **88-92%**[72].

🟦 ARDS (Acute Respiratory Distress Syndrome)

- **Definition**: Alveolar-capillary membrane damage leading to "stiff lungs" (decreased compliance)[73][73][73][73].
- 🎯 **Hallmark**: **Refractory Hypoxemia** (PaO2 remains low despite high FiO2)[74].
- **Treatment**: PEEP (Positive End-Expiratory Pressure) and Prone Positioning[75].

🟦 Tuberculosis (TB)

- **S/S**: Persistent cough (>3 weeks), Night sweats, Weight loss, Hemoptysis[76].
- **Isolation**: 🎯 **Airborne Precautions** (Negative pressure room). Staff wear **N95**[77].
- **Diagnosis**: Sputum Culture (Acid-fast bacillus) is definitive[78].

🟦 Emergencies (PE & Trauma)

- **Pulmonary Embolism (PE)**: Sudden dyspnea, "Impending Doom". Dx: **CT Angiography**[79].
- **Tension Pneumothorax**: Tracheal deviation to unaffected side, JVD, hypotension. Tx: **Needle Decompression**[80].
- **Flail Chest**: Paradoxical Chest Movement[81].

🟪 VII. Pharmacology

🟦 Drug Table

Class	Drug	Indication	Critical Nursing Intervention
TB Meds (RIPE)	**Rifampin**	TB	Turns body fluids **Red-Orange**. Do not wear contacts[82].
	Isoniazid (INH)	TB	🎯 Risk of Neuropathy. Administer **Vitamin B6** to prevent[83].
	Ethambutol	TB	Risk of Ocular Toxicity (Report vision changes)[84].
Bronchodilators	**Albuterol**	Asthma/COPD	Rescue inhaler. Monitor for Tachycardia/Tremors[85].
	Theophylline	Asthma	⚠️ Narrow therapeutic index (10-20 mcg/mL). Toxic: Seizures[86].
Emergency	**Succinylcholine**	Intubation	Trigger for **Malignant Hyperthermia**[87].

Antidotes	**Dantrolene**	MH	The **ONLY** antidote for Malignant Hyperthermia[88].
	Acetylcysteine	Mucolytic	Smells like rotten eggs. Antidote for **Acetaminophen**[89].

🟪 VIII. Case Study: ADPIE Application

📝 **Scenario**: Mr. Jones, 68, Post-Op Day 2 from abdominal surgery. Smoker. Complaining of SOB[90].

🟩 Assessment

- **Data**: RR 28, SpO2 89% on Room Air. Breath sounds diminished in bases with fine crackles. Anxious[91].

🟩 Diagnosis

- Ineffective Airway Clearance r/t retained secretions and pain[92].
- Impaired Gas Exchange r/t alveolar collapse[93].

🟩 Planning/Outcome

- Patient will maintain SpO2 > 92% and demonstrate correct use of Incentive Spirometer within 1 hour[94].

🟩 Implementation

- 🩺 **Positioning**: High Fowler's to maximize lung expansion[95].
- 🩺 **Oxygen**: Apply nasal cannula at 2 L/min[96].
- 🩺 **Action**: Administer pain meds, then assist with **Splinting** and **TCDB**. Encourage Incentive Spirometer (10x/hr)[97].

🟩 Evaluation

- SpO2 improves to 95%. Patient produces sputum. Breath sounds clear[98].

🟪 IX. Must Know (Exam High-Yield)

🎯 Agitation/Restlessness

The #1 early sign of Hypoxia/Hypoxemia99.

🎯 Stridor

A medical emergency indicating upper airway obstruction100.

🎯 Rifampin Side Effect

Causes red-orange discoloration of body fluids; this is normal101.

🎯 Silent Chest

In asthma, no wheezing indicates complete obstruction and impending arrest102.

🎯 Tracheal Deviation

Confirmatory sign of Tension Pneumothorax requiring needle decompression103.

🎯 Paradoxical Chest Movement

The hallmark of Flail Chest104.

🎯 Hemorrhage Alert

Chest Tube Drainage > 100 mL/hr is excessive105.

🎯 Pink Frothy Sputum

Classic sign of severe Pulmonary Edema106.

🎯 Refractory Hypoxemia

Key diagnostic feature of ARDS (low PaO2 despite high O2)107.

🎯 Beta-Blockers Contraindication

Non-selective beta-blockers are contraindicated in Asthma/COPD (causes bronchospasm)108.

🎯 Continuous Bubbling

In the water seal chamber indicates an air leak; tidaling is normal109.

🎯 Malignant Hyperthermia

Triggered by Succinylcholine. Treat with Dantrolene110.

🎯 Metformin Safety

Withhold 48 hours before/after contrast dye procedures to prevent lactic acidosis111.